Homework 6

Perturbation theory and tight-binding.

1. (70 pts) We will return to the last problem set to use perturbation theory. Consider the same potential as last time for the square lattice:

$$V(\vec{r}) = \sum_{m_1,m_2} V_{(m_1,m_2)} e^{\mathrm{i}(m_1\vec{b}_1\cdot\vec{r}+m_2\vec{b}_2\cdot\vec{r})}$$

$$V_{(1,0)} = V_{(\bar{1},0)} = V_{(0,1)} = V_{(0,\bar{1})} = -0.04 \qquad V_{(1,1)} = V_{(\bar{1},\bar{1})} = V_{(\bar{1},1)}V_{(1,\bar{1})} = -0.02$$

(a) (10 pts) Given the equation we derived in class for the first order correction, does this potential give a first order correction? Why or why not?

(b) (20 pts) Write a k-dependent analytic equation for the *second order* corrections to the lowest energy branch (ie. the $(0,0)$ branch) along the k-path $(0,0) \to (\frac{1}{2},0)$.

(c) (15 pts) Plot the analytic equation derived in the previous part, along with the free value and the value you obtained by diagonalizing the 9x9 matrix in the previous problem set (ie. the lowest value eigenvalue along that k-path).

(d) (25 pts) You will see that you are having problems in $\vec{k}_\ell = (\frac{1}{2},0)$ because of the degeneracy of the bare levels. The essence of degenerate perturbation theory is to make a unitary transformation to a better basis such that the degeneracy is eliminated. Let us do this for the entire branch. Start by writing the 2x2 matrix for the $\vec{k}+(0,0)$ and $\vec{k}+(\bar{1},0)$ states, and write the analytic eigenvalue for the lower branch. Now write down the second order corrections to this new branch. As above, please plot the free branch, the result of the 9x9 matrix, the analytic eigenvalue for the lower branch of the 2x2 matrix you just found, and finally the included 2nd order correction to the previous curve.

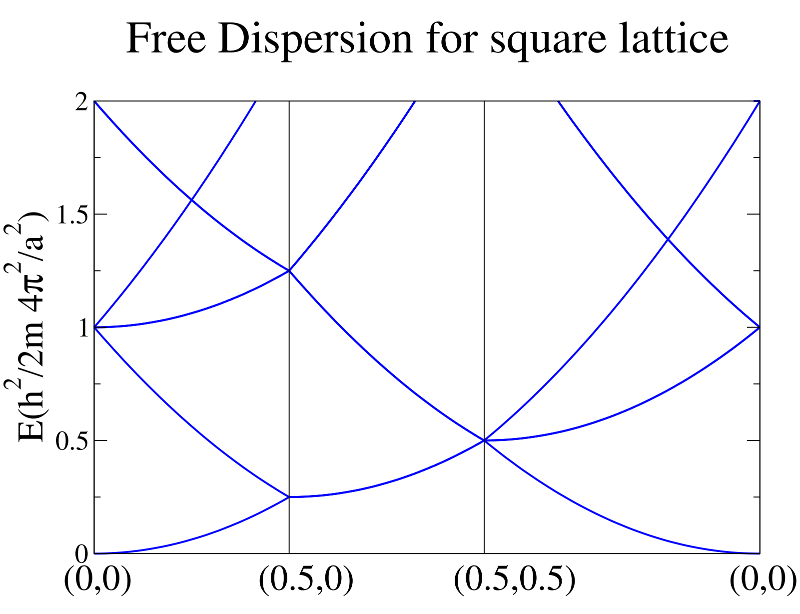

Figure 1: Free electrons for the 2d empty square lattice with parameter a. The k-point path is indicated in the x-axis in lattice coordinates (ie. fractions of the reciprocal lattice vectors $\vec{b}_1$ and $\vec{b}_2$).

2. (30 pts) Consider a one dimensional lattice with some potential. Let us assume a localized basis set with one hydrogen s-orbital on each lattice site. Let us assume that there is only nearest neighbor hopping t. Now assume that we apply a perturbation such that $\langle\phi_i|\hat{H}|\phi_i\rangle = \epsilon_o + \epsilon(-1)^i$, where $\epsilon << t$ and i labels a site in the unperturbed lattice.
   (a) (25 pts) Find the analytic solution for the band structure, and plot the results assuming that $t = 1$ and $\epsilon = 0.05t$.
   (b) (5 pts) If we have one electron per site, is the system a metal or an insulator?